California Property Close Price Prediction

Task Prompt & 12-Week Intern Game Plan

Task Prompt: Predicting California Property Close Price (Final Sales)

Accessing the Datasets

1. Download FileZilla Client from https://filezilla-project.org
2. Establish FTP connection using Host: 199.250.207.194, Username: data@idxexchange.com, Password - Real_estate123$, Port: 21 (Type in the password instead of copying and pasting).
3. Datasets are in raw/California and have the prefix ‘CRMLSSold’.
4. Please make a copy of the ‘CRMLSSold’ files and download to your local machine, and do not modify any of the files in the raw folder. Download a minimum of 6 months of data.
5. Document containing the meta data for the datasets is ‘Trestle Property MetaData.pdf’ and is in the ‘resources’ folder.
6. Please use only observations with PropertyType="Residential" and PropertySubType="SingleFamilyResidence" when constructing the model.

Background

You have been provided with datasets containing historical close price (final sales price) data on real estate properties sourced from CRMLS (California Regional Multiple Listing Service), including various features such as living area, number of bedrooms, bathrooms, lot size, and close prices. ClosePrice is the target variable — the value the model is being trained to predict, using the property’s other characteristics as input features. Your task is to develop a machine learning model to predict the close price of any single residential property (currently for sale or not for sale) in California based on its characteristics at the time of the query.

Task Description

1. Data Exploration: Begin by exploring the dataset to understand its structure, features, and any patterns present in the data. Identify relevant features that could influence the close price of a property.
2. Data Preprocessing: Preprocess the dataset as needed, including handling missing values, encoding categorical variables, and scaling numerical features if necessary. Split the data into training and testing sets for model evaluation.
3. Model Selection: Choose an appropriate machine learning model for predicting property close prices. You may start with a simple model like linear regression or explore more complex models such as decision trees, random forests, or gradient boosting.
4. Model Training: Train the selected model using the training data. Tune hyperparameters if applicable to optimize model performance.

5. Model Evaluation: Evaluate the trained model using appropriate metrics such as R-squared on the testing data. Assess the model's performance and identify areas for improvement.
6. Prediction: Once the model is trained and evaluated, use it to predict the close price of a particular property based on its characteristics. Provide the predicted close price as the output of the model.
7. Documentation: Document the entire process, including data exploration findings, preprocessing steps, model selection rationale, training methodology, evaluation results, and the prediction process.

Deliverables

- Python script containing the code for data preprocessing, model training, evaluation, and prediction.
- Documentation detailing the steps taken during the analysis and reasoning behind decisions made at each stage.
- Live presentation via Zoom summarizing key findings, model performance, and the prediction process for stakeholders.

Additional Notes

- Ensure that the code is well-organized, commented, and follows best practices for reproducibility and readability.
- Experiment with different features, models, and hyperparameters to improve prediction accuracy.
- Seek feedback from team members or domain experts to validate the model's effectiveness and refine the approach if necessary.

12-Week Intern Game Plan for California Price Prediction

Week 1 — Orientation & Setup

- Read the Task Prompt doc and understand the goal.
- Install Python, Git, IDE, and and/or Google Collab.
- Establish FTP connection (FileZilla) to download a minimum of 6 months of CRMLSSold files from /raw/California.
- Review Trestle Property MetaData.pdf in /resources to understand feature definitions.
- **Deliverable:** Confirm dataset access; write a short note on what each key column means.

Week 2 — Data Exploration

- Load a minimum of 6 months of dataset into pandas.
- Explore distributions of ClosePrice (the target variable), LivingArea, Bedrooms, Bathrooms, LotSize.
- Restrict analysis to PropertyType = Residential and PropertySubType = SingleFamilyResidence (per task doc).
- **Deliverable:** Jupyter notebook 01_exploration.ipynb with basic EDA plots.

Week 3 — Data Preprocessing

- Handle missing values (decide whether to drop, impute, or flag).
- Convert categorical fields to numeric (encoding).
- Normalize numerical features if needed.
- Create train/test split: use the most recent month of available data as the test set, and the X months immediately preceding it as the training set. X is not fixed — treat the training window length as a tunable choice and experiment to determine the optimal value of X.
- **Deliverable:** 02_preprocessing.ipynb + cleaned CSV.

Week 4 — Baseline Model

- Train a Linear Regression as the first model.
- Evaluate using R^2 on the test set.
- Record baseline results.
- **Deliverable:** 03_baseline_model.ipynb.

Week 5 — Additional Models

- Try Decision Tree and Random Forest regressors.
- Compare their test R^2 against baseline.

- Document model behavior (strengths/weaknesses).
- **Deliverable:** 04_model_comparison.ipynb.

Week 6 — Feature Engineering

- **Example of sample features you can engineer:** bed/bath ratio, age of property in years
- **Adding more detailed geographic layer using school districts:** build a more detailed regional feature by spatially joining each property's coordinates against the CA School District Areas 2024-25 boundaries (https://data.ca.gov/dataset/california-school-district-areas-2024-25/resource/7dfaf005-58eb-45db-93b1-7aff091b2172)
- Re-train models with the updated feature set.
- **Deliverable:** Updated notebook + table comparing old vs new feature sets, including the school district layer.

Week 7 — Advanced Models

- Try Gradient Boosting (e.g., XGBoost or LightGBM).
- Perform light hyperparameter tuning (depth, learning rate, n_estimators).
- **Deliverable:** 05_advanced_models.ipynb with test metrics.

Week 8 — Evaluation Expansion

- Compute metrics beyond R^2: MAPE and MdAPE.
- Summarize insights (e.g., which price bands perform better).
- **Deliverable:** 06_evaluation.ipynb + metrics_summary.csv.

Week 9 — OPTIONAL Simple Prediction App (Streamlit)

- Build a Streamlit app: user inputs LivingArea, Beds, Baths, LotSize → output predicted price.
- Load trained model with joblib/pickle.
- **Deliverable:** app.py (streamlit app).

Week 10 — Documentation

- Write a README describing: dataset source, preprocessing, models tested, best results.
- Document instructions to re-run the code and launch the app.
- **Deliverable:** README.md.

Week 11 — Practice Presentation

- Prepare a slide deck summarizing: data, methods, models, evaluation, Streamlit demo.

- Rehearse with peers or mentors.
- **Deliverable:** Slide draft.

Week 12 — Final Team Presentation & Handoff

- Deliver final Zoom presentation to stakeholders (arrange presentation time with Aidan).
- Submit final repo with: Python scripts (preprocessing, training, evaluation, prediction), Documentation (README, metadata notes), App (app.py).
- **Deliverable:** Final repo + slides + live demo.
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